

Methodological issues in the development of a national database for primary care groups and trusts

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Abstract

At the National Primary Care Research and Development Centre (NPCRDC) we have constructed a national database for all primary care groups (PCGs) in England. At its core, the database links information about population socio-economic and demographic characteristics to generic health status and to the organisation, resourcing and activities of general practice. In this paper we describe and discuss the problems with linking these data, and with defining the boundaries and the local populations of PCGs, given that they have been established on the basis of administrative expediency rather than geographical coherence. We then consider the implications of these difficulties for needs assessment in primary care groups.

Keywords: healthcare geography, needs assessment, primary care database, primary care groups/trusts

Primary care groups

The formation of primary care groups (PCGs) in England in April 1999 marked the dawning of a new era in the planning and provision of primary care. The central focus of these groups is to commission services for patients in order to 'secure appropriate, high quality care for local people', with particular emphasis on improving public health and the coordinated provision of primary care, through closer collaboration between health and social services (Department of Health 1998, 1999). Primary care groups have grown out of a number of recent initiatives taken in the National Health Service, that have involved general practices in commissioning services for their patients. For example, GP fundholding, in which a GP practice managed a budget of its practice staff, certain hospital referrals, drug costs, community nursing services and management costs. Locality commissioning groups were another model in which groups of general practices worked closely with their health authorities to plan and commission services in a particular area. Previous models for commissioning services in the primary care setting have now been superseded by primary care groups in England. In Scotland the equivalent is the primary care trust and in Wales, the local health group.

The PCG is essentially composed of a group of general practices and their registered patients, the average size of the population served by these groups being approximately 100 000. Each PCG is run by a committee composed, for the most part, of general practitioners, but also including representatives of other providers of primary care (e.g. nurses, social services) and lay members. Each PCG is responsible to its local health authority. There are four possible 'levels' of responsibility that PCGs can take in relation to the provision of primary care for their populations. At a minimum (level 1) they can simply support the health authority in commissioning care for its population, acting in an advisory capacity. Level 2 PCGs take devolved responsibility for managing the budget for health-care in their area, formally as part of the health authority. At levels 3–4 PCGs can move towards becoming primary care trusts (PCTs), establishing themselves as free-standing bodies accountable to the health authority for commissioning care (level 3) and taking on the responsibility for the provision of community health services for their population (level 4). But at whatever level PCGs operate, they share with other organisations in the NHS the broad goals of providing fair access to care in relation to population need, of ensuring that care is delivered effectively and is appropriate, and of working toward

health improvement and reducing health inequality (Department of Health 1998, 1999, National Health Service Executive 1998a).

These broad aims are laudable, but pragmatically monitoring the progress of PCGs in achieving these goals will be difficult to achieve since there is currently a lack of comparative national data that enables a systematic link to be drawn between primary health and social care provision, population characteristics and health outcomes for locality-based populations. Existing databases such as the National Morbidity Statistics (McCormick *et al.* 1996) focus on the morbidity of those consulting the general practitioner and publish no information about the organisation, and activities of primary care that have been so closely associated with the health of locality-based populations. They are also practice- rather than locality-based, using samples of general practices that are self-selecting, so that all localities are not equally represented. Similar problems exist for national surveys such as the General Household Survey (Office for National Statistics 1999) and the Health Survey of England (Colhoun & Prescott-Clarke 1994), for which only a small number of cases are likely to be located in each PCG and these will probably not be representative of the PCG population. Hence, at the National Primary Care Research and Development Centre we have set up a web-based information system – The National Database for Primary Care Groups/Trusts – to provide the linked data that is so vital for NHS managers and policy-makers to assess the achievements of PCG/Ts (www.primary-care-db.org.uk). It also provides an information resource for researchers in the field of primary care, epidemiology, public health and health policy.

In building this database, our first task was to provide baseline descriptions of the socio-economic and demographic characteristics of PCG populations and patterns of primary health-care provision. Now this 'core' database is established, we intend to add to it more detailed information about health need, morbidity and mortality, about prescribing, and about the organisation and provision of social and community services. We have also constructed a map-based interface to the database, so that data are easily accessible by area.

What at first sight may appear to be the relatively straightforward task of linking data about population characteristics and the provision of primary care by locality is complicated by the configuration of the relevant data sets, which are based on different 'geographies' and are collected at contrasting time periods, in the early and the late 1990s. Also, there were no nationally coherent geographical boundaries for PCG/Ts, and these required definition, but the information available to do this was incomplete. Both these factors affect the

way that PCG/T populations are estimated and, as a consequence, the assessment of population need.

The linkage of different geographies

Information about the characteristics of local populations, including age, sex, ethnicity, occupational groups, unemployment, and other proxies for population need is most comprehensively provided by the 1991 census of the population. The smallest areas for the management of enumeration in the 1991 census are enumeration districts (EDs). These are typically made up of between 150 and 250 households. The 107 000 EDs used in 1991 aggregate directly into 9000 wards, which are administrative populations of approximately 5000 people. These, in turn, form part of a hierarchical structure that builds up into districts, counties and standard regions.

The biannually updated General Medical Services Statistics for England (GMS) have information about the staffing of general practice, levels of activity and primary care provision, and age and sex of registered practice populations for all of the 9000 practices in England. We receive GMS data by general practice and by general practitioner. Each practice and each GP is allocated a unique identifying code, which can be linked to an address and a postcode, using an address look-up table. The geographical location of each practice is thus the practice postcode. A unit postcode represents a collection of mail delivery points, it does not represent an area with clearly and consistently defined boundaries. Although postcodes build hierarchically into larger areas, such as postcode sectors, 'postcode' geography bears no systematic relationship to the administrative geographies employed in the 1991 census (Denham 1993).

The 1998 NHS Postcode Directory (a truncated version of the All Fields Postcode Directory) was used as an integrating framework for the different geographies, and the postcode was used as the building block for the database. The postcode directory provides grid references for each unit postcode, based on the location of its first address; A PCG code number is, in principal, attached to each postcode, but in practice some health authorities were slow to provide this information. Consequently, there were some information gaps in the database, which have only very recently been resolved. A 1991 ward code was also derived from the postcode directory and so it was possible to locate each GP practice in a 1991 census ward and aggregate these wards into their respective PCGs. This method provides a reasonably good estimate of PCG population characteristics and the provision of primary care, but there are some measurement issues that may affect the precision

of the postcode to ward link. For example, postcodes are constantly changing and so the accuracy of the database is dependent on the correctness of practice postcodes given in GMS statistics (Barr 1993a, 1993b). It is sometimes thought that postcode to enumeration district is a more appropriate means of linking data, as the latter are smaller units that aggregate directly into wards. However, postcodes frequently fall into two or more EDs, posing complex problems of allocation by postal address (Reading & Openshaw 1993, Gatrell *et al.* 1991). The postcode to ward link is not so problematic, because wards are bigger area units, although potentially problems can arise at or near ward boundaries, with postcodes being assigned to an adjacent ward by mistake. Another problem with using postcodes as the building block for the database is that national grid references are widely acknowledged to be spatially imprecise. At the time of writing, existing systems are about to be replaced by a new product called Grid-Link. This has been developed jointly by The Office for National Statistics (ONS), Consignia and Ordnance Survey, and links all UK postcodes to standard administrative and health areas. Postcodes will have one-metre resolution grid references to identify the postcode centroid, which will be clipped to an address within the postcode. Grid-Link will undoubtedly improve the quality of postcode geography, but it is expensive to implement, and will not solve the problem of poor or outdated information at source.

PCG boundary definitions

One fundamental problem with the construction of PCG boundaries was that specification of their definition from the Office of National Statistics (ONS) was somewhat at odds with guidelines appearing in key National Health Service Executive (NHSE) policy documents. ONS requested that health authorities constitute their PCGs in terms of whole current (1998) electoral wards. Theoretically, this would allow the existing postcode to electoral ward links to be used to create the appropriate postcode to PCG link (personal communication, Alan Taylor, ONS). But the directives given to health authorities regarding the configuration of PCGs by the NHSE were sufficiently vague to allow for liberal interpretation. For example PCGs 'should embrace natural communities that make sense to the local populations and allow inequalities to be addressed, as well as meeting the needs of the mobile (homeless, travellers and refugee) population' (National Health Service Executive 1998a, p. 8), or that the configuration should be 'geographically congruent with the distribution of any minority group which would be advantaged by a relationship with one PCG' (National Health Service Executive 1998a, p. 8).

Consequently, only 77 of the 100 health authorities constituted their PCGs according to the ONS requirement. The remainder have defined their PCGs on the basis of 'mix and match' geographies. Eighteen health authorities constituted some of their PCGs using whole current electoral wards, and the remainder in terms of other geographies, such as split current electoral wards, 1991 census wards and 1991 census enumeration districts. The remaining five health authorities supplied their entire constitution in terms of other geographies, such as 1991 census enumeration districts, and unit postcodes or postcode sectors.

The PCGs that are defined on the basis of this 'mix and match' geography will inevitably have boundaries that are at best 'fuzzy' – their exact location cannot be precisely determined. For example, 1998 electoral wards may well be split between one or more PCGs. Where this occurs, boundaries have had to be inferred using postcodes as the mapping tool. In such cases, we have apportioned the ward population according to the number of residential delivery points that fall into each of the PCGs in question.

Where health authorities have defined their PCG boundaries in relation to 1991 census wards and enumeration districts, or postcode areas, there are likely to be both areas of overlap and 'black holes', where these PCGs border PCGs using current geographies.

The difficulties outlined above suggest that no definitive set of PCG boundaries can currently be drawn. 'Fuzzy' boundaries are, however, likely to become more clearly defined in the future, as PCGs merge into PCTs. PCTs are statutorily required to define their boundaries using current ward geographies, although ward definitions will continue to change over time, and boundaries require adjustment as a consequence. To keep the ward/PCG link up-to-date, it may be necessary to refer back to 1991 EDs. These have the same 'code-ends' as the ward in which they are contained and this should allow the appropriate link to be made.

Despite the problems that we have encountered, hard or fuzzy boundaries have been defined for all 481 PCGs in England. We now go on to consider how defining PCG populations in different ways affects the calculation of socio-economic and demographic indicators of population need.

Primary care group populations and needs assessment

Resident versus registered populations

A primary care group population can be estimated using either the resident or the registered population. The 'resident' population is defined as those people

who live within the geographical boundaries of a PCG. The resident population of each PCG has been calculated on the national database by aggregating ward level data from the 1991 census. Populations are given by sex and in 5-year age groups. More up-to-date projections of resident PCG populations for 1999 have been produced by ONS and are available from the Public Health Common Data Set (PHCDS) (Balarajan *et al.* 1999). These are based on mid-1991 estimates from the census, making allowance for area-specific birth, death and migration rates and population ageing.

Registered PCG populations are all those people registered with the general practices that make up a PCG. We have obtained these data from GMS statistics and they are available on the database by age-group (age-groups are 0–1, 2–4, 5–15, 16–24, 25–44, 45–64, 65–74, 75+) and sex for 1997, 1998 and 1999. Potentially, this is a more appropriate PCG population base than the resident population, because PCGs were set up to serve those people registered with a GP within their group. However, populations derived from practice lists tend to be inflated, due to lags in the removal of patients who have moved or who have died (National Health Service Executive 1998a, 1998b).

A third source of information about PCG populations is based on the resident/registered population overlap – the proportion of the registered population resident within PCG boundaries. This information is issued to health authorities for the purposes of resource allocation, and is available on the Attribution Data Set. Unfortunately, this data set is not yet available for dissemination on the database.

Comparing population denominators

We compared the two resident and the registered PCG populations to ascertain the possible effects of using different population denominators for PCG needs assessment. We used the information on all 481 PCGs from the national database and compared the three different population bases, both in absolute and relative terms. We then made the same comparison for the four PCGs within Manchester Health Authority, for which there is additional information about resident populations drawn from a local 1996 census (Manchester Health Authority 1999).

Figure 1 shows that there is a very strong correlation between the three population bases; pairwise Pearson and Spearman-rank correlations are all greater than 0.9. Thus, relative to one another, we would expect PCG constitution to be very similar, regardless of which base population is used.

As expected, when we considered the absolute size of PCG populations, registered populations were generally

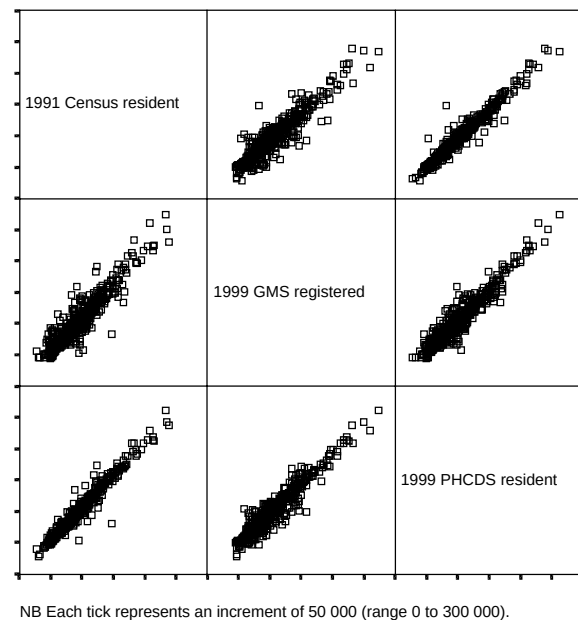


Figure 1 Matrix scatterplot of base populations.

larger than resident populations ($\bar{x}_{\text{census_resident}} \cong 96\,500$; $\bar{x}_{\text{GMS_registered}} \cong 105\,800$; $\bar{x}_{\text{PHCDS_resident}} \cong 101\,900$), and therefore PCG constitution may be substantially affected by the choice of population. The graphs show that there are a number of large deviations from the line of equality.

The consistently greater size of the registered populations is probably mainly the consequence of list inflation. For PCGs in Manchester Health Authority we were able to measure the extent of list inflation (Table 1), and this shows considerable variation across PCGs in the same health authority. This factor varies from a high of 17.2% in Manchester East to a low of 9.4% in Manchester South. Table 1 also shows that the PHCDS-projected resident populations for Manchester PCGs are lower than those quoted in the health authority report, and, with the exception of West PCG, lower than those obtained from the 1991 census. This leads to some doubt about the reliability of these estimates and it is uncertain as to whether these are replicated across the country.

Estimating indices of PCG population 'need'

But how are these differences in the estimated size of PCG populations likely to affect the measurement of population need? This depends on which measures are being used to approximate need. For demographic characteristics such as the percentage of elderly in the PCG or the percentage of under-fives (groups known to have a high need for primary care services) the most up-to-date estimates could be derived for registered

PCG	Resident population			Registered population	
	HA Report*	Census†	PHCDS‡	HA Report§	GMS¶
Manchester North	109 610	104 390	98 765	124 933	124 232
Manchester South	128 291	120 167	114 264	146 324	140 373
Manchester East	107 636	104 387	94 117	127 736	126 192
Manchester West	82 153	66 868	73 087	97 232	94 047

* Mid-1997 estimate, based on 1996 local census.

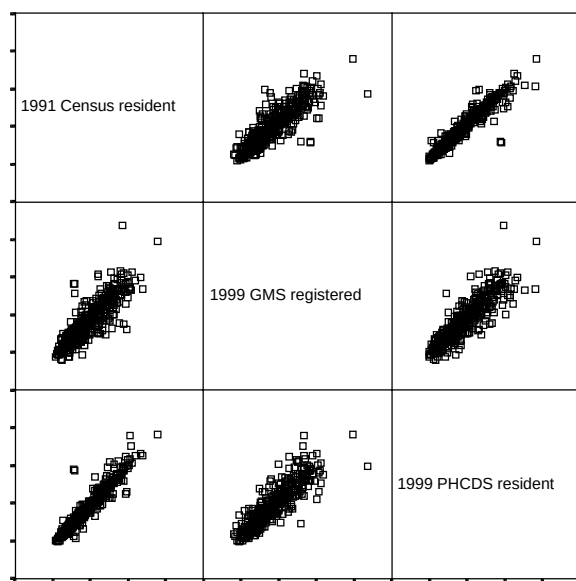
† Based on 1991 census data.

‡ ONS 1996-based projections (to 1999).

§ Based on March 1998 practice registers.

¶ Based on October 1999 GMS statistics data.

Table 1 A comparison of resident and registered populations from the NPCRDC PCG Database and the 1999 Manchester Public Health Annual Report

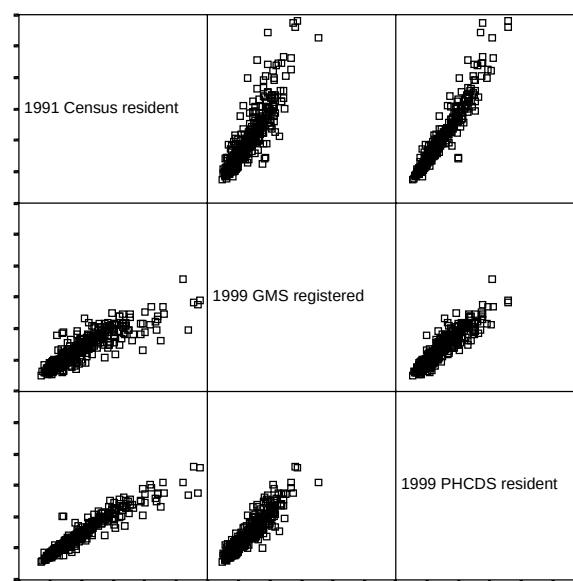


NB Each tick represents an increment of 50 (range 0 to 250).

Figure 2 Matrix scatterplot of limiting long-term illness rates.

populations. The accuracy of these estimates would, however, be improved by adjustment using a PCG level list inflation factor.

The estimation of other proxies for need such as levels of unemployment, ethnicity and limiting long-term illness are more problematic if registered populations are used as the denominator, because need will be underestimated in these cases. For example, Figure 2 illustrates directly age-standardized rates of limiting long-term illness (LLTI) per 1000 population, calculated using each of our three base populations. The pairwise correlations between the three sets of LLTI rates are very high, but the rates calculated using either of the resident populations are generally greater than those using the 1999 registered populations. This is a direct result of the registered populations being larger than the resident populations. There were no LLTI rates



NB Each tick represents an increment of 5% (range 0 to 30).

Figure 3 Matrix scatterplot of unemployment rates.

available for comparison in Manchester Health Authority PCGs.

Added to this potential for underestimation of need is the problem of definition of socio-economic variables such as unemployment. Figure 3 illustrates unemployment rates, calculated using the different base populations. The obvious feature of this set of graphs is that unemployment rates calculated using the 1991 resident populations are greater for all but a handful of PCGs. This could be due to the other two population bases being larger; it could also reflect a genuine fall in unemployment in the 1990s, but it could also arise because of different definitions of unemployment. The definition of unemployment used in the 1991 census was the percentage of all economically active residents who were unemployed, whereas only those people of working age can be estimated using the 1999 registered

Table 2 Male unemployment rates for Manchester PCGs

PCG	Population base			
	1991 resident population*	1999 registered population†	1999 PHCDS population‡	1998 ONS estimates§
North	24.0	14.6	19.8	12.0
South	19.0	11.6	15.5	11.0
East	25.5	14.1	20.4	15.9
West	26.1	12.3	15.9	17.6

* Percentage of economically active males who are unemployed.

† Percentage of males aged 16–64 who are unemployed.

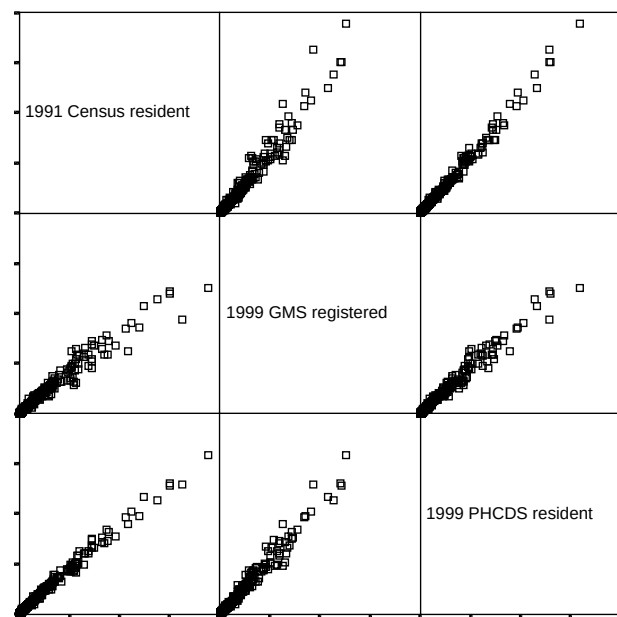
‡ Percentage of males aged 15–64 who are unemployed.

§ Percentage of the eligible male workforce who are unemployed and claiming benefit.

populations and the resident PHCDS populations. Clearly, the 'economically active' are a subset of the population of working age and will not include those who have retired early, are permanently sick, or students, thus decreasing the number 'at risk' of being unemployed.

Table 2 shows male unemployment rates for Manchester PCGs, calculated using the definitions above, and, additionally 1998 ONS estimates. The latter are defined as the percentage unemployed and claiming benefit. These rates are most similar to those calculated using the 1999 registered populations, but are considerably lower than the rates calculated from the 1991 resident population. With the exception of Manchester West, they are also lower than those calculated using the 1999 PHCDS populations. It is difficult to offer any coherent interpretation of the reliability of these various rates of unemployment, because of the different definitions used. One plausible alternative in measuring need associated with poor socio-economic circumstances is to use the new ward-based indices of income and unemployment produced by the Department of the Environment, Transport & the Regions – DETR (2000) that are now available on the database. Ward level population estimates produced by the Office for National Statistics have been used as the population denominator for these measures, but the unreliability of these estimates needs to be offset against the particular problems associated with the use of census data.

Two further problems that can arise when estimating the needs of PCG populations are well illustrated by considering the ethnic composition of PCGs. This is calculated from the 1991 census as the percentage of the PCG population who belong to an ethnic minority group. The vast majority of PCGs have a very small proportion of their population who are classified as ethnic minorities (census median = 2.08%, registered median = 1.96%, PHCDS median = 1.99%). Figure 4 thus gives the impression that the ethnic composition of most PCGs is similar, regardless of which base population is used in the calculation. It is only in PCGs with a



NB Each tick represents an increment of 20% (range 0 to 80).

Figure 4 Matrix scatterplot of ethnic composition.

higher ethnic composition that the use of a different population base becomes apparent. For example, Southall and West Ealing PCG had 47 996 residents according to the census, 56 769 residents according to the PHCDS and 71 929 registered patients. A total of 36 336 individuals were classified as members of ethnic minority groups in the 1991 census; the ethnic composition of Southall and West Ealing PCG would therefore be 75.7% using the census data, 64.0% using the PHCDS data, but only 50.5% using the GMS registered populations.

Table 3 shows the ethnic minority composition of Manchester PCGs. The fourth column of data is recorded as being obtained from the 1991 census, and one might expect that it would agree exactly with that calculated from the census on the PCG database. Small differences are, however, evident. These probably arise

PCG	Population base			
	1991 Resident population	1999 Registered population	1999 PHCDS population	1999 Annual Report*
North	8.6%	7.2%	9.1%	8.5%
South	6.7%	5.7%	7.0%	6.5%
East	14.0%	11.6%	15.5%	14.0%
West	29.2%	20.8%	26.8%	28.1%

* Based on the 1991 census.

Table 3 Ethnic minority composition of Manchester PCGs

from different apportioning of the Central ward resident population that is split between North, East and West PCGs.

To summarize, these analyses show that 1999 registered and 1999-projected resident PCG populations are good approximations of 1991 resident PCG populations on a relative basis. Moreover, the Manchester Health Authority data suggest that, in terms of absolute size, the registered populations may well be reasonable estimates of more recent (1996) resident populations. Currently, list inflation factors are only calculated at health authority level. If list inflation factors were available at PCG level, 1999 registered populations could provide reasonably precise population 'denominators' to estimate demographically based needs indices such as those derived from age and sex.

We also conclude that it would not be wise to use registered populations to derive PCG-level measures of need based on 1991 census variables such as ethnicity, unemployment and limiting long-term illness. Absolute numbers are likely to be distorted due to differences between the population denominators in both size and definition. Similar conclusions apply to the use of the resident populations from the Public Health Common Data Set. Absolute differences between these populations and those from the census are not as great (see Fig. 1), and any differences may well just be a function of time. However, the calculation of absolute levels of need using certain indicators, such as unemployment rates, will still be distorted by different definitions of the population 'at risk'. The 1999-projected data for Manchester PCGs also appeared to underestimate populations, a pattern that could be mirrored in other PCGs across England. A plausible alternative to measuring need associated with poor socio-economic circumstances is to use the new ward-based indices produced by the Department of the Environment, Transport and the Regions. However census data will remain the only source of information for a number of key needs' indicators, for example ethnicity and limiting long-term illness. For these variables our view is that the resident 1991 population should be employed as the population denominator.

Concluding comments: the viability of a national database for primary care groups and trusts

The essential value of The National Database for Primary Care Groups/Trusts will be in the longitudinal study of change in primary care provision and its relation to improvement in population health and reductions in health inequality. Cross-sectional comparative studies of patterns of health and primary care in PCGs with similarly disadvantaged populations will also be important. Accurate estimation of contrasting sources, and levels of population need is an obvious prerequisite for such research.

What has become clear in constructing the core components of this database is that a coherent geographical basis for linking data from different sources and for defining the boundaries of administrative areas is necessary in order to make such an information system viable. At present, as has been illustrated in this paper, the linkage of data sets based on different geographies has meant that PCG populations can be defined in relation to those people registered at the GP practices within the PCG, or those resident within the administrative boundaries of the PCG. The appropriate population base is dependent on the particular variables that are the focus of analysis and their availability at the practice or the area level (Bardsley *et al.* 2001). The problem of matching data sets based on different administrative geographies will become even more complicated when building data on social service provision into the database, since currently only 23 PCGs are coterminous with the local authority areas for which such data are routinely collected. It is to be hoped that as PCGs merge into PCTs, their boundary definitions will match more completely with existing administrative areas.

Another major issue that affects the viability of the national database and its potential for development is that of the quality, and the extent, of available information. As we have shown, key variables for needs assessment, such as ethnicity and limiting long-term illness, are currently only available for the whole of England from the 1991 census. The 2001 census is currently underway,

but the reliance on a decennial source of information about population need will necessarily compromise the conclusions that can be drawn about the impact of changes in primary care on the outcomes of care for locality-based populations. In addition, much of the information that is of direct relevance for such research is currently restricted to NHS use, which limits the potential of the national database for creating a comprehensive information system and consistently hampers the production of high quality evaluative studies by academic departments and non-NHS users.

The 'core' database is now available for academic and NHS users via its own website. Currently it is structured as a secure information system from which data can be downloaded at a number of levels of aggregation. The process of updating and adding new data sets to the 'core' is ongoing and we have recently completed an interactive map-based interface for easy access to the data, using MAPGUIDE®. Such GIS software provides a unique two-way link between data and map. It contains a sophisticated database that stores attribute data referring to points, lines and areas of the map. Once the map has been plotted, any point, line or area may be highlighted, and information relating to it may be queried.

One important challenge for the future development of the national database for primary care groups and trusts will be to keep abreast of boundary changes caused by mergers of PCGs into PCTs and by administrative adjustments to postcodes and to wards. This will ensure the provision of a coherent longitudinal data set for analysis of change over time. Another challenge is to provide data analysis tools for our users. The provision of information is not always, by itself, enough to enable users of the database to identify and to analyse material that is relevant to their particular area of interest.

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